Phase Timeline

Project Start: 2026-05-27 Target Completion: 2026-07-01 (5 weeks)

Phase 1: Architecture & Research (Week 1)

Objective: Define system architecture and select technical stack

Key Activities: - Research procurement workflows (interview someone in procurement/operations) - Compare agent frameworks (LangGraph, AutoGen, CrewAI, n8n) - Design memory architecture (supplier profiles, decision history) - Define decision logic framework (scoring algorithm, constraint checking) - Document ethical safeguards and risk mitigation strategies

Deliverables: - Framework selection decision (with justification) - Memory system design document - Decision logic specification - Risk assessment and safeguard plan - Decision log entry: Architecture choices

Success Criteria: - Clear rationale for framework choice - Memory system handles all required data types - Decision logic addresses edge cases - Safeguards cover identified risks

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Phase 2: Core Autonomous Loop - Manual Simulation (Week 2)

Objective: Build and test basic detect -> decide -> act workflow with manual data

Key Activities: - Create mock inventory data with stockout scenarios - Implement threshold detection logic - Build supplier evaluation system (hardcoded options initially) - Generate purchase order format - Implement human approval workflow - Test with 3-5 simulated scenarios

Deliverables: - Working prototype (manual triggers) - Mock data generators - PO generation capability - Test scenario results - Decision log entry: Initial implementation approach

Success Criteria: - Agent detects stockout conditions correctly - Supplier evaluation logic runs without errors - PO format matches business requirements - Human can approve/reject with clear reasoning visible

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Phase 3: Memory & Learning Integration (Week 3)

Objective: Add memory system and enable learning from decisions

Key Activities: - Implement supplier history tracking - Build supplier scoring system (performance-based) - Add pricing trend analysis - Create learning loop (decision -> outcome -> score update) - Test that second decision improves based on first outcome - Implement delivery tracking and delay detection

Deliverables: - Memory system integration - Supplier scoring algorithm - Learning feedback loop - Comparative test results (1st vs 2nd decision quality) - Decision log entry: Memory architecture implementation

Success Criteria: - Supplier scores update after transactions - Agent makes better decisions with more history - Memory persists across sessions - Pricing trends influence future decisions

Phase 4: Edge Cases & Robustness (Week 4)

Objective: Test failure modes and implement recovery strategies

Key Activities: - Test supplier out-of-stock scenario - Test delivery delay handling - Test price spike detection - Implement duplicate invoice detection - Test budget constraint violations - Create escalation workflows for unhandled cases

Deliverables: - Edge case test suite - Recovery strategy implementations - Escalation workflow documentation - Invoice reconciliation system - Decision log entry: Edge case handling strategies

Success Criteria: - Agent handles all identified edge cases gracefully - No silent failures (all errors logged/escalated) - Duplicate invoice detection catches test cases - Budget constraints prevent overspend

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Phase 5: Safeguards, Ethics & Demo Prep (Week 5)

Objective: Implement safety measures and prepare final demonstration

Key Activities: - Implement financial limits (max autonomous spend) - Build audit trail system (decision reasoning logs) - Conduct bias analysis (supplier selection fairness) - Create human override mechanisms - Test with realistic end-to-end scenarios - Document ethical considerations - Prepare demonstration materials

Deliverables: - Complete safeguard implementation - Audit trail documentation - Ethics section for decision log - Final demonstration scenarios - Complete decision log with all phases documented - Presentation materials

Success Criteria: - Agent meets all success metrics (2+ stockout prevention, 1+ invoice error caught, 95%+ approval) - Safeguards prevent catastrophic decisions - Audit trail enables human understanding of all decisions - Ethics documentation addresses transparency, accountability, fairness, safety, privacy - Demo shows both happy path and edge case handling